

```

-- Transaction
-- start transaction;
-- begin;

-- savepoint savepointName;
-- rollback to savepointName;

-- rollback;

-- commit;


-- How to create a new user:-
create user 'username'@'host' identified by 'password';

-- host concept:-
-- host          meaning
-- localhost      only local machine
-- %              Any machine
-- 192.168.1.10   Specific IP


create user 'shamal'@'localhost' identified by 'shamal@123';
create user 'shamal'@'%' identified by 'shamal@123';
create user 'shamal'@'192.168.1.10' identified by 'shamal@123';


-- checking permissions:-
show grants for 'username'@'hostname';


-- Privileges level:-
-- 1) global level privilege :-
on *.*;

-- 2) Database level:-
on database_name.*

-- 3) Table level:-
on database_name.table_name

-- 4) Column level:-
grant select(emp_id, first_name, department)
on amazon.employees
to 'shamal'@'localhost';


-- DCL :-
-- data control language:-

```

```
-- DCL is used to control access and permissions on database objects
-- DCL decides who can do what in your database.
```

```
-- two main dcl commands in mysql
-- grant and revoke
```

```
-- 1)Grant:-
-- grant is used to give permissions to users.
-- syntax
GRANT privilege_TYPE
ON DATABASE_NAME.TABLE_NAME
TO 'USERNAME'@'HOST';
```

```
GRANT SELECT, INSERT
ON UDFS.GPAY_CUSTOMERS
TO 'shamal'@'localhost';
```

```
grant update
on udfs.gpay_customers
to 'shamal'@'localhost';
```

```
-- how to see grants(privileges) for a specific user
show grants for 'shamal'@'localhost';
```

```
-- common privlieges
/*
select      read data
insert      add new data
update      modify data
delete      remove data
all privileges      full access
create      create database/table
drop        drop database/tables
alter       modify table structure
*/
```

```
grant all privileges
on udfs.*
to 'shamal'@'localhost';
```

```
grant all privileges
on *.*
to 'shamal'@'localhost';
```

```
-- creating a admin user
create user 'admin1'@'localhost'
```

```
identified by 'admin123';
```

```
-- with grant option:-  
-- used to give grant option to other users so that they can grant privileges to  
-- other users
```

```
-- note (*.*) means all_databases.all_tables)  
grant all privileges  
on *.*  
to 'admin1'@'localhost'  
with grant option;
```

```
-- 2) Revoke:-  
-- used to remove permissions from a user  
-- syntax  
revoke privilege_type  
on database_name.table_name  
from 'username'@'hostname';
```

```
-- example  
-- revoking shamal's insert and update privileges from  
-- udfs database and gpay_customer table  
revoke insert, update  
on udfs.gpay_customers  
from 'shamal'@'localhost';
```

```
-- how to revoke all privileges also revoking grant option  
revoke all privileges, grant option  
from 'admin'@'localhost';
```

```
-- create connection for user john  
create user 'john'@'localhost'  
identified by 'john123';
```

```
-- HOW TO SEE USER AND HOST FOR THOSE USER IN MYSQL  
SELECT USER, HOST FROM MYSQL.USER;
```

```
-- CREATE A USER WITH YOUR OWN NAME ON LOCALHOST  
-- ESTABLISH A CONNECTION FOR THAT USER  
-- AND GIVE PRIVILEGES FOR ANY ONE DATABASE AND ALL THE TABLES INSIDE THAT DATABASE  
-- PRIVILEGES SHOULD BE FOR INSERTING, UPDATING AND SELECTING DATA  
-- THEN REVOKE UPDATING AND INSERTING FOR THAT USER.
```

```
-- 1ST TASK  
-- GIVE SELECT PRIVILEGE OF COMPANY_SP_PRACTICE.EMPLOYEES  
-- FOR THE COLUMN  
-- EMP_ID, EMP_NAME AND AGE
```

```
-- 2ND TASK
-- GIVE ALL PRIVLIGES FOR DEPARTMENTS TABLE IN C_S_P DB TO THE USER
-- INSERT A NEW DEPARTMENT INSIDE A TRANSACTION
-- AND COMMIT THOSE CHANGES.
```